

Allan Vikiru

Data Science | Applied AI | Internet of Things

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Career Profile

- Over 4 years of experience in progressively responsible roles in data science, artificial intelligence and internet-of-things.
- Demonstrated strengths in enhancing human-computer interaction by building sustainable systems including a project leveraging AI and IoT in the fight against illegal logging.
- Knowledgeable in languages for statistical modelling like Python, R and SQL for machine learning and spatial data analysis.
- Holder of a Master's in Information Technology (Business Intelligence and Data Analytics).

Skills and Competencies

- | | | |
|-------------------------|---|--------------------------|
| - Software Engineering | - Data Engineering | - Technical Writing |
| - Machine Learning | - Web and Cloud Development | - Communication Skills |
| - Hardware Prototyping | - Object-Oriented Programming | - Project Management |
| - Spatial Data Analysis | - Commercial and Academic Presentations | - Stakeholder Engagement |
| - IoT Prototyping | | |

Languages: C++, JavaScript, LATEX, MATLAB, Python, R

Software: Arduino, InfluxDB, Git, Django, Heroku, Microsoft Power BI, QGIS, Streamlit

Work Experience

Tutorial Fellow - School of Computing & Engineering Sciences

Strathmore University

July 2024 - Present

Role: University Teaching; Research; Supervision.

Achievements:

- Teaching at least 250 students on Artificial Intelligence, Embedded Systems, IoT, Information Security, Data Governance and Cloud Computing.
- Coordinating research activities at the School's Makerspace Lab including engaging at least 10 industry players and research Institutions, and 40 undergraduate and master's students.
- Supervising up to 25 students on projects related to challenge-driven education, deep learning and Internet of Things.

Research Coordinator - School of Computing & Engineering Sciences (SCES) Makerspace Lab

Strathmore University

June 2023 - Present

Role: Research; Research Coordination

Achievements:

- Continually engaging a team of 6 researchers and at least 10 undergraduate and graduate students in developing computing and engineering projects.
- Co-lead of the **Conservation Technology Research Hub**. Key tasks included internal and external stakeholder engagement, data collection and analysis, device prototyping, grant proposal writing and conducting team meetings.

Graduate Assistant - School of Computing & Engineering Sciences

Strathmore University

March 2022 - June 2024

Role: Research; Proposal Writing; Research Coordination; Stakeholder Engagement; Project Management; Field Testing.

Achievements:

- Developed a research proposal on using machine learning and IoT to detect logging activities that received an award of USD 139,500.

- Contributor to three research papers on deep learning and cloud computing.
- Created a health informatics project involving an algorithm that uses spatial soil data to recommend foods for diabetic patients, which registered 80% accuracy.

Software Developer Intern

@iLabAfrica, IoT & Wireless Networks Lab

January 2022 - March 2022

Role: Software Development; Stakeholder Engagement

Achievements:

- Designed and implemented a monitoring dashboard in Vue.js and Node.js and an authentication system in Keycloak for a LoRaWAN-based research project.
- Designed and developed breadboard model devices to model LoRaWAN connectivity.
- Engaged at least 50 stakeholders in the planning and disseminating workshops on the Internet of Things and spectrum management.

Education

MSc. Information Technology (Business Intelligence & Data Analytics Option)

Strathmore University - Nairobi, Kenya

2024

BSc. Informatics & Computer Science

Strathmore University - Nairobi, Kenya

2021

Conferences & Workshops

Delegate – GDH Partner Days on Challenge Driven Education

Royal Institute of Technology (KTH) - Stockholm, Sweden

March 2026

Presenter - A Wearable Sensor System for Monitoring the Risk of Fireground Injuries among Firefighters

11th International Congress on Information and Communication Technology - London, United Kingdom

February 2026

Delegate – Microsoft Faculty Skilling and Immersion Program (Cohort 3 – AI-Oriented Education)

Microsoft ADC & Microsoft Leap - Nairobi, Kenya

June 2025

Delegate - 5th International Conference on Smart Sensors and Application

Universiti Teknologi Malaysia (UTM) - Penang, Malaysia

September 2024

Project Administrator - Curbing Illegal Logging Patterns Using Sound-Based Detection Techniques

Strathmore University (Kenya), Université de Neuchâtel (Switzerland), University of Huddersfield (United Kingdom), Kenya Forestry Research Institute (Kenya)

July 2023 - September 2024

Delegate – CMU Innovation Summit 2024

Carnegie Mellon University Africa - Kigali, Rwanda

January 2024

Presenter – Fundamentals of Raspberry Pi 400

Strathmore University - Nairobi, Kenya

November 2023

Delegate - International Summer School in Cyber Security and Digital Forensics

Brno University of Technology - Brno, Czech Republic

June 2023

Publications

Vikiru A. & Khakata E. (in press). A Wearable Sensor System for Monitoring the Risk of Fireground Injuries among Firefighters.

11th International Congress on Information and Communication Technology.

Ayankoso, S., Wang, Z., Shi, D., Yang, W., **Vikiru, A.**, Kamau, S., Muchiri, H., & Gu, F. (2024). Development of Long-Range, Low-Powered and Smart IoT Device for Detecting Illegal Logging in Forests. *Journal of Dynamics, Monitoring and Diagnostics*.

<https://doi.org/10.37965/jdmd.2024.550>

Simiyu, D., **Vikiru, A.**, Muchiri, H., Gu, F., & Butime, J. (2024). A Chainsaw-Sound Recognition Model for Detecting Illegal Logging Activities in Forests. In A. D. Ball, H. Ouyang, J. K. Sinha, & Z. Wang (Eds.), *Proceedings of the UNified Conference of DAMAS, InCoME and TEPEN Conferences (UNified 2023)* (pp. 797–806). Springer Nature Switzerland. [https://doi.org/10.1007/978-3-](https://doi.org/10.1007/978-3-031-49421-5_65)

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Vikiru, A. (2023, March 18). *Market Basket Analysis using Apriori Algorithm*. <https://allanvikiru.github.io/MarketBasketAnalysis/>

Vikiru, A., Muiruri, M., & Ateya, I. (2023). *An Overview on Cloud Distributed Databases for Business Environments*

(arXiv:2301.10673). arXiv. <https://doi.org/10.48550/arXiv.2301.10673>